

NUTRIMITRA: PERSONALIZED DIET MANAGEMENT SYSTEM FOR INDIA
A PROJECT REPORT

Submitted in partial fulfilment of the Requirements for the award of the Degree of BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)
By [Student Name]
[Roll Number]
Under the esteemed guidance of [Guide Name] (Assistant Professor)

DEPARTMENT OF INFORMATION TECHNOLOGY
[College Name] (AUTONOMOUS) (Affiliated to University of [University])
[City], [Pincode]
[State]
[Academic Year]

ABSTRACT
Nutrimitra is a comprehensive diet management mobile application designed to help users achieve their nutritional goals through personalized diet plans and meal suggestions.

ACKNOWLEDGEMENT
I express my sincere gratitude to Principal [Name], Coordinator [Name] for their continuous support and encouragement throughout the project.

DECLARATION
I hereby declare that the project entitled "Nutrimitra: Personalized Diet Management System for India" is my original work done and submitted by [Student Name].

TABLE OF CONTENTS

Sr. No.	Contents	Pg. No.
Chapter 01: Introduction		
1.1	Background	01
1.2	Objective	02
1.3	Purpose	02
1.4	Scope	03
1.5	Applicability	04
1.6	Achievements	04
1.7	Organization Report	05
Chapter 02: Survey Of Technology		
2.1	Introduction	06
2.2	Front-End Technology	06
2.2.1	React Native	06
2.2.2	Expo Sdk	07
2.2.3	TypeScript	07
2.2.4	User Interface Libraries	07
2.3	Back-End Technology	08
2.3.1	Firebase Authentication	08
2.3.2	Firebase Cloud Firestore	08
2.3.3	Firebase Cloud Storage (Optional)	09
2.3.4	Google ML Kit (Food Recognition)	09
2.3.5	Nutrition API Integration	09
2.4	Device And Platform Apis	09
2.5	Development Tools And Environment	10
Chapter 03: Requirement And Analysis		
3.1	Project Definition	11
3.2	Requirement Specification	12
3.2.1	Functional Requirement	12
3.2.2	Non-Functional Requirement	13
3.3	Planning And Scheduling	14
3.4	Gantt Chart	14
3.5	Hardware Requirements	15
3.6	Software Requirements	15

3.7 Preliminary Product Description	15
3.8 Modules Description	16
3.9 Feasibility Study	17
Chapter 04: System Design	18
4.1 Diagrams	18
4.1.1 Activity Diagram	18
4.1.2 Class Diagram	19
4.1.3 Use Case Diagram	20
4.1.4 Entity-Relationship Diagram	21
4.1.5 Data Flow Diagram	22
4.2 Data Design	23
4.2.1 Schema Design	23
4.2.2 Data Integrity and Constraints	24
4.3 Algorithm Design	25
4.4 User Interface Design	26
4.5 Test Case Design	28
Chapter 05: Implementation And Testing	30
5.1 Implementation And Approaches	30
5.1.1 Implementation	30
5.1.2 Approaches	33
5.2 Coding Details And Efficiency	34
5.3 Testing Approach	35
Chapter 06: Result And Discussion	38
6.1 Test Report	38
6.2 User Documentation	39
6.2.1 Authentication Module	39
6.2.2 Food Logger Module	40
6.2.3 Meal Planner Module	41
6.2.4 Nutrition Tracker Module	42
6.2.5 Water Intake Module	43
6.2.6 Recipe Suggestion Module	44
6.2.7 Health Condition Module	45
6.2.8 Nearby Nutrition Services Module	46
6.2.9 Settings & Preferences	47
6.3 Observations	48
Chapter 07: Conclusion	49
7.1 Conclusion	49
7.2 Limitations Of The System	49
7.3 Future Scope Of The Project	50
References	51

CHAPTER 01 - INTRODUCTION

1.1 Background

India faces a dual burden of malnutrition where undernutrition coexists with rising rates of obesity, diabetes, and cardiovascular

1.2 Objectives

The major objectives of the Nutrimitra project are:

1. To develop a mobile application featuring an extensive Indian food database with nutritional values for regional dishes, ingredients, and recipes.
2. To create personalized meal planning functionality that considers user health conditions (diabetes, hypertension, weight goals) and dietary preferences.
3. To implement food intake tracking with macro/micro nutrient analysis using image recognition and manual entry methods.
4. To provide water intake monitoring with intelligent reminders based on individual needs and environmental factors.
5. To offer recipe suggestions with healthy Indian cooking methods and ingredient substitutions.
6. To integrate with wearable devices for synchronized activity and sleep data to refine nutritional recommendations.
7. To implement location-based services for finding nutritionists, diet clinics, healthy restaurants, and organic stores.
8. To support multi-language interface enabling users to interact in their preferred Indian language.
9. To ensure secure data handling with persistent user sessions and privacy protection.
10. To promote sustainable healthy eating habits through educational content and progress visualization.

1.3 Purpose of the Project

The purpose of Nutrimitra is to empower individuals to take control of their nutritional health through technology that understands

1.4 Scope of the Project

Nutrimitra encompasses the following core components:

- Indian Food Database: Curated nutritional information for 5000+ Indian foods including regional variations, street foods, thalis,
- Personalized Meal Planning: Dynamic plan generation based on health profiles (weight loss, muscle gain, diabetes management)
- Food Tracking System: Multiple input methods (image recognition, voice, barcode, manual search) with nutrient breakdown (calories, macros, vitamins)
- Hydration Tracker: Intelligent water reminders adjusted for climate, activity level, and individual physiology
- Nutrition Analytics: Visual reports showing trends, deficiencies, and goal progress with actionable insights
- Recipe Engine: Healthy Indian recipes with step-by-step instructions, nutritional info, and ingredient alternatives
- Health Condition Modules: Specialized guidance for conditions like diabetes, PCOD, thyroid, and pregnancy
- Local Services Integration: Location-based discovery of nutrition professionals, health food stores, and diet-friendly restaurants
- Multi-Language Support: Interface available in Hindi, Bengali, Tamil, Telugu, Marathi, Gujarati, Kannada, Malayalam, Punjabi,
- Wearable Sync: Integration with popular fitness trackers for holistic health view

1.5 Applicability

Nutrimitra serves diverse user groups across India:

1. Health-Conscious Individuals: Seeking to optimize nutrition for fitness goals or general wellbeing
2. Patients with Lifestyle Conditions: Managing diabetes, hypertension, or cholesterol through diet
3. Pregnant and Lactating Women: Requiring stage-specific nutritional guidance
4. Elderly Population: Addressing age-related nutritional needs and deficiencies
5. Fitness Enthusiasts: Athletes and gym-goers needing precise macro tracking
6. Students and Professionals: Busy individuals requiring quick, healthy meal options
7. Nutritionists and Dietitians: Tool for client management and personalized plan delivery
8. Corporate Wellness Programs: Employee health initiatives focusing on nutrition
9. Public Health Organizations: Community nutrition programs and awareness campaigns
10. Food Industry: Restaurants and food brands seeking to provide nutritional transparency

1.6 Achievements of the Project

Key milestones achieved during Nutrimitra development:

- Created comprehensive Indian food database with 5000+ items verified against NIN and ICMR standards
- Implemented meal planning algorithm that generates culturally appropriate Indian meal plans
- Developed food recognition model trained on Indian dishes achieving 85% accuracy for common foods
- Built hydration tracking system with contextual intelligence (weather, activity-based adjustments)
- Integrated multi-language support enabling seamless switching between 10 Indian languages
- Implemented secure authentication with biometric option and session persistence
- Designed offline-first architecture for core tracking functionality with sync capability
- Developed visualization dashboard showing nutrient trends and goal compliance
- Created recipe modification engine for healthier versions of traditional Indian dishes
- Implemented location services for finding nutritionists and health food outlets in major cities

1.7 Organization of the Report

This report follows a structured approach detailing all project aspects:

- Chapter 1 - Introduction: Project background, objectives, scope, and significance
- Chapter 2 - Survey of Technologies: Analysis of frameworks, libraries, and tools selected
- Chapter 3 - Requirement and Analysis: Functional/non-functional requirements, feasibility study, module planning
- Chapter 4 - System Design: Architectural decisions, data models, algorithms, and interface design
- Chapter 5 - Implementation and Testing: Development approach, coding details, and validation methods
- Chapter 6 - Result and Discussion: Testing outcomes, user documentation, and system observations
- Chapter 7 - Conclusion and Future Scope: Project summary, limitations, and enhancement plans
- References: Technical sources, APIs, and research materials consulted

CHAPTER 02 - SURVEY OF TECHNOLOGIES

2.1 Introduction

Selecting appropriate technologies was critical for Nutrimitra's success as a nutrition-focused mobile application serving diverse

2.2 Front-End Technology

Nutrimitra's user interface is built using React Native with Expo SDK to deliver a consistent experience across Android and iOS

2.2.1 React Native

React Native enables cross-platform mobile development using JavaScript and React, providing near-native performance through

- Single Codebase: Maintain one codebase for both Android and iOS, reducing development overhead
- Native UI Components: Ensures familiar interaction patterns for Indian smartphone users
- Performance: Sufficient for food logging and tracking operations without lag
- Community Support: Extensive libraries available for nutrition-specific features
- Over-the-Air Updates: Enables quick deployment of database updates and feature improvements

2.2.2 Expo SDK

Expo simplifies React Native development by providing managed workflow and access to device capabilities:

- Pre-configured Environment: Eliminates complex native setup for camera, location, and sensors
- Expo Go App: Facilitates instant testing on physical devices during development
- OTA Updates: Critical for pushing Indian food database updates without app store delays
- Built-in Libraries: Includes modules for notifications (meal reminders), location (nearby services), and file system (offline data s
- Web Support: Enables future expansion to progressive web app for desktop access

2.2.3 TypeScript

TypeScript enhances code quality and maintainability for Nutrimitra's complex nutritional logic:

- Type Safety: Prevents errors in nutrient calculation and meal planning algorithms
- IDE Support: Better autocompletion for food database schemas and API integrations
- Refactoring Confidence: Safer modifications to core nutrition logic as features evolve
- Documentation: Self-documenting code through explicit type definitions
- Scalability: Supports growing codebase as more regional foods and health conditions are added

2.2.4 User Interface Libraries

To create an intuitive interface for diverse Indian users, these libraries were employed:

- @expo/vector-icons (MaterialIcons): Consistent iconography recognizing Indian dietary symbols
- react-native-paper: Material Design implementation optimized for Indian color preferences
- react-native-responsive-screen: Ensures proper layout across diverse Indian device screens
- react-native-svg: Enables custom visualizations for traditional Indian meal plate representations
- react-native-picker-select: Culturally appropriate selection controls for language and region choices

2.3 Back-End and Cloud Services

Nutrimitra utilizes Firebase as its backend infrastructure for reliable data handling and user management.

2.3.1 Firebase Authentication

Secure user management is essential for personal nutritional data:

- Email/Password Authentication: Primary login method with secure password handling
- Google Sign-In: Convenient option integrated with existing Android ecosystem
- Phone Number Authentication: Important for users preferring mobile-based verification
- Anonymous Mode: Allows exploration before commitment, respecting privacy concerns
- Session Management: Persistent login through secure tokens handling intermittent connectivity
- Custom Claims: Enables role-based access for nutritionists vs regular users

2.3.2 Firebase Cloud Firestore

Firestore's document model suits Nutrimitra's hierarchical nutritional data:

- User Profiles: /users/{uid} storing health goals, conditions, preferences, and metrics
- Food Logs: /users/{uid}/foodLogs/{date} with meal-wise entries and nutrient totals

- Meal Plans: /users/{uid}/mealPlans/{planId} containing daily/weekly Indian meal suggestions
- Recipes: /recipes/{recipeId} with ingredients, instructions, and nutritional info per serving
- Water Logs: /users/{uid}/waterLogs/{date} tracking intake with contextual reminders
- Progress Records: /users/{uid}/progress/{metric} storing weight, measurements, and health markers
- Real-Time Sync: Instant updates across devices when users modify plans or log meals
- Offline Persistence: Core functionality available without internet, syncing when reconnected

2.3.3 Firebase Cloud Storage (Optional)

For future enhancements involving media:

- Food Images: User-uploaded meal photos for community recipe sharing
- Recipe Media: Step-by-step cooking videos for traditional Indian preparations
- Educational Content: Infographics on Indian nutrition concepts and myth busting
- Security Rules: Ensures users only access their own media or approved public content

2.3.4 Google ML Kit (Food Recognition)

On-device machine learning for practical food logging:

- Image Classification: Identifies common Indian foods from meals (dal, roti, rice, vegetables)
- Custom Model: Trained on 10,000+ Indian food images across regional variations
- Portion Estimation: Approximates serving size based on plate context and known dish types
- Barcode Scanning: Fallback for packaged foods with integrated Indian product database
- Privacy-First Processing: All image analysis occurs on device without uploading photos
- Fallback Mechanism: Manual search when recognition confidence is low

2.3.5 Nutrition API Integration

External services for comprehensive nutritional data:

- USDA FoodData Core: Base nutritional values augmented with Indian-specific data
- Indian Food Composition Tables (IFCT): Region-specific nutrient profiles from NIN
- Packaged Food Database: Branded products common in Indian markets with labeling data
- Regional Variants: Adjustments for North/South/East/West Indian preparation methods
- Fasting/Festival Foods: Special nutritional data for religious observances and fasts

2.4 Device and Platform APIs

Nutrimitra leverages device capabilities through Expo-managed APIs:

- expo-camera: For food image capture and recognition
- expo-location: To find nearby nutritionists, diet clinics, and healthy food outlets
- expo-notifications: Smart reminders for meals, water intake, and meal prep times
- expo-sensors: Step count and activity data from phone sensors for calorie adjustment
- expo-av: Audio guidance for cooking traditional Indian recipes
- expo-secure-store: Encrypted storage for sensitive health data
- expo-blur-view: Privacy protection for meal logging in public settings
- AsyncStorage: Local caching of food database and user preferences for offline use

2.5 Development Tools and Environment

- Visual Studio Code: Primary IDE with extensions for React Native and TypeScript development
- Node.js (LTS): Runtime for executing JavaScript tooling and dependency management
- npm/yarn: Package management for frontend libraries and development scripts
- Git/GitHub: Version control with branching strategy for feature development
- Expo CLI: Project scaffolding, development server, and build automation
- Firebase Console: Backend configuration, monitoring, and rules management
- Android Studio/Xcode: For native module testing and device-specific debugging
- Postman: Testing nutrition API integrations and Firebase functions
- Figma: UI/UX design and prototyping for Indian user preferences

CHAPTER 03: REQUIREMENT AND ANALYSIS

3.1 PROBLEM DEFINITION

Achieving optimal nutrition in India faces distinct challenges:

1. Nutritional Information Gap: Lack of accessible, accurate data on Indian foods' nutritional content
2. One-Size-Fits-All Advice: Generic diet plans ignoring regional preferences, economic constraints, and cultural practices
3. Tracking Complexity: Difficulty measuring nutrients in traditional Indian meals with mixed dishes and variable portions
4. Behavioral Barriers: Forgetting to log meals, inconsistent tracking, and lack of motivation without feedback
5. Limited Professional Access: Inadequate availability of qualified nutritionists, especially in rural areas
6. Seasonal Variations: Changing food availability and nutritional needs across Indian seasons and festivals
7. Health Condition Management: Difficulty adapting traditional diets for conditions like diabetes or hypertension
8. Hydration Neglect: Insufficient attention to water intake despite its critical role in nutrition
9. Language Barriers: Nutrition information predominantly in English excluding vernacular speakers
10. Lack of Contextual Intelligence: Apps not adjusting recommendations based on activity, climate, or health status

3.2 REQUIREMENT SPECIFICATION

Nutrimitra specifies requirements to deliver effective, culturally relevant nutrition management:

3.2.1 FUNCTIONAL REQUIREMENTS

1. User Profiling

- Capture health metrics (weight, height, age, gender, activity level)
- Record health conditions (diabetes, hypertension, PCOD, pregnancy status)
- Set nutritional goals (weight loss, muscle gain, maintenance, therapeutic)
- Specify regional food preferences and restrictions (vegetarian, vegan, Jain, etc.)
- Store language preference for interface localization

2. Food Database & Recognition

- Maintain searchable database of 5000+ Indian foods with nutritional values per standard Indian serving
- Implement image-based food identification using device camera
- Support barcode scanning for packaged foods with Indian labeling
- Allow manual entry with predictive text for common Indian dishes
- Enable saving custom foods and recipes with user-defined nutritional info

3. Meal Planning & Suggestions

- Generate daily/weekly Indian meal plans based on user profile and goals
- Provide alternative options for each meal to accommodate taste preferences
- Adjust plans for fasting days, festivals, and seasonal availability
- Suggest ingredient substitutions for healthier versions of traditional dishes
- Integrate with local seasonal produce availability data

4. Nutrition Tracking

- Log breakfast, lunch, dinner, and snacks with time stamps
- Calculate macro nutrients (protein, carbs, fat) and micro nutrients (fiber, sugar, sodium, vitamins)
- Display nutrient intake vs. goals with visual progress indicators
- Flag potential deficiencies or excesses based on established guidelines (ICMR, WHO)
- Provide meal quality scores based on food diversity and processing levels

5. Hydration Management

- Calculate personalized water needs based on weight, activity, climate, and health conditions
- Schedule intelligent reminders throughout waking hours
- Log water intake with container size recognition (glass, bottle, traditional vessels)
- Adjust recommendations based on exercise and weather data
- Show hydration trends and consistency metrics

6. Recipe & Cooking Guidance

- Offer healthy Indian recipes with step-by-step instructions

- Provide nutritional information per serving for all recipes
- Suggest cooking methods that preserve nutrients (steaming, sautéing vs deep frying)
- Offer ingredient alternatives for common allergens or restrictions
- Include traditional preparation methods with modern health adaptations

7. Health Condition Modules

- Diabetes: Carbohydrate counting, glycemic index guidance, meal timing
- Hypertension: Sodium tracking, DASH diet adaptations using Indian ingredients
- Weight Management: Calorie deficit/surplus planning with sustainable Indian meals
- Pregnancy: Trimester-specific nutritional needs with Indian food sources
- Elderly: Easy-to-chew options, nutrient-dense foods for common deficiencies
- Pediatric: Age-appropriate nutrition with Indian complementary foods

8. Location-Based Services

- Find registered nutritionists and dietitians within specified radius
- Locate health food stores, organic markets, and farmers' markets
- Identify restaurants offering healthy Indian options with nutritional info
- Locate pharmacies for supplements when medically recommended
- Map walking trails and parks for activity complementing nutrition

9. Multi-Language Support

- Interface available in 10 major Indian languages (Hindi, Bengali, Tamil, Telugu, Marathi, Gujarati, Kannada, Malayalam, Punjabi)
- Allow seamless language switching without losing progress
- Provide culturally relevant examples and references in each language
- Support voice input and output in selected languages
- Ensure nutritional terms are accurately translated and locally understood

10. Data Synchronization & Privacy

- Secure authentication with options for biometric login
- Persistent sessions handling intermittent connectivity
- Local storage for core functionality during offline periods
- Encrypted synchronization when network available
- User control over data sharing and deletion rights
- Compliance with Indian data protection principles

3.2.2 NON-FUNCTIONAL REQUIREMENTS

1. Performance

- App launch time under 3 seconds on mid-range Indian smartphones
- Food logging completion under 15 seconds per meal
- Nutrient calculation and display within 2 seconds of food entry
- Smooth scrolling and interaction on devices with 2GB RAM
- Battery consumption optimized for all-day use with background tracking

2. Security & Privacy

- All personal health data encrypted at rest and in transit
- Biometric authentication option (fingerprint/face ID)
- No nutritional data shared without explicit user consent
- Option to use app completely offline with local storage
- Clear data retention and deletion policies
- Secure handling of sensitive health condition information

3. Scalability

- Support for 10,000+ concurrent active users
- Efficient handling of growing food database (target: 10,000+ items)

- Smooth operation with extensive user history (2+ years of data)
- Ability to add new regional cuisines and health condition modules
- Cloud functions for complex nutritional calculations when needed

4. Usability & Accessibility

- Interface designed for users with varying literacy levels
- Large touch targets and clear visual hierarchy
- Voice navigation for eyes-free operation during cooking
- Color schemes accessible to users with visual impairments
- Works effectively in low-light kitchen environments
- Tutorials and help available in all supported languages

5. Maintainability

- Modular architecture separating UI, logic, and data layers
- Well-documented APIs and data schemas
- Automated testing coverage for critical nutrition calculations
- Clear separation between core engine and region-specific content
- Version-controlled nutritional database updates

6. Cultural Relevance

- Accurate representation of diverse Indian cuisines (North, South, East, West, Central)
- Respect for dietary restrictions based on religion, region, and tradition
- Inclusion of traditional Indian superfoods and medicinal foods
- Festival and seasonal food guidance integrated naturally
- Recognition of traditional measurement systems (katori, cup, handful, etc.)

3.3 PLANNING AND SCHEDULING

Development followed a phased approach to ensure thorough implementation:

- Phase 1: Foundation & Core Database (4 weeks)
 - Indian food database creation and verification
 - Basic user authentication and profile management
 - Core food logging interface with manual entry
- Phase 2: Tracking & Recognition (5 weeks)
 - Image-based food recognition implementation
 - Nutrient calculation engine and meal logging
 - Water intake tracking with smart reminders
- Phase 3: Planning & Personalization (4 weeks)
 - Meal planning algorithm based on health goals
 - Recipe integration and suggestion system
 - Health condition-specific modules
- Phase 4: Localization & Expansion (3 weeks)
 - Multi-language interface implementation
 - Location-based services for nutrition professionals
 - Wearable device integration
- Phase 5: Refinement & Launch (4 weeks)
 - User testing with diverse Indian demographic groups
 - Performance optimization and battery efficiency
 - Documentation and deployment preparation
 - Feedback incorporation and final polishing

3.4 GANTT CHART

Project timeline visualized through Gantt chart showing:

- Parallel work streams for frontend and backend development
- Overlapping testing phases with development milestones

- Buffer periods for integration and unexpected challenges
- Milestone reviews at end of each major phase
- Final two weeks dedicated to user acceptance testing and polishing

3.5 HARDWARE REQUIREMENTS

- Development Machines: Intel i5 processor, 8GB RAM, 256GB SSD
- Testing Devices: Range of Android (versions 8-14) and iOS (versions 13-17) devices
 - Minimum: 2GB RAM, quad-core processor, 8GB storage
 - Recommended: 3GB RAM+, octa-core processor, 16GB+ storage
- Peripherals: Standard webcam for video consultation features (future)
- Network: Stable broadband connection for development and testing
- Target Users: Any smartphone with Android 8.0+ or iOS 13.0+ (covers 95% of active Indian devices)

3.6 SOFTWARE REQUIREMENTS

- Operating Systems: Windows 10/11, macOS Ventura+, Ubuntu 22.04 LTS
- Development IDE: Visual Studio Code with React Native and TypeScript extensions
- Frontend Stack: React Native 0.72+, Expo SDK 49, TypeScript 5.0+
- Backend Services: Firebase Authentication, Firestore, Cloud Functions (Blaze plan)
- AI/ML: Google ML Kit (Vision API), TensorFlow Lite for custom food model
- Nutrition Data Sources: USDA FoodData Central, IFCT 2017, NIN databases
- Mapping Services: OpenStreetMap via Leaflet/Mapbox for location features
- Version Control: Git 2.30+, GitHub for collaboration and CI/CD
- Testing: Jest for unit testing, Detox for end-to-end, Firebase Test Lab for device matrix
- Design: Figma for UI/UX, Adobe Illustrator for nutritional graphics
- Documentation: Markdown for technical docs, MkDocs for publication

3.7 PRELIMINARY PRODUCT DESCRIPTION

Nutrimitra functions as a personalized nutrition companion that:

1. Learns user preferences, health status, and goals over time
2. Provides context-aware meal suggestions fitting Indian eating patterns
3. Tracks actual consumption against planned intake with visual feedback
4. Educates users about nutritional content of their traditional foods
5. Adapts recommendations based on activity, health metrics, and goals
6. Connects users with local nutritional resources when needed
7. Supports sustainable habit formation through gradual improvement focus
8. Respects India's culinary diversity while promoting nutritional adequacy

3.8 MODULES DESCRIPTION

1. Authentication & Profile Management

- Secure login with email/phone/social options
- Health profile creation with goals, conditions, and preferences
- Regional food preference setup (North/South/East/West Indian, vegetarian types)
- Language selection for interface localization
- Biometric login option for convenience and security

2. Indian Food Database & Recognition

- Searchable food database with 5000+ items categorized by type, region, and meal course
- Image capture and processing for food identification
- Barcode scanning for packaged foods with Indian labeling support
- Manual entry with intelligent suggestions for common dishes
- Custom food and recipe creation with user-defined nutrition facts

3. Meal Planning Engine

- Daily/weekly plan generation based on caloric/macronutrient targets

- Health condition adaptations (diabetes, hypertension, etc.)
- Regional cuisine preferences integration
- Festival and seasonal adjustment capabilities
- Grocery list generation from meal plans
- Leftover incorporation suggestions to reduce waste

4. Nutrition Tracker & Analyzer

- Meal logging with time-stamped entries for breakfast, lunch, dinner, snacks
- Automatic nutrient calculation using food database
- Macro/micro nutrient breakdown with goal comparison
- Visual intake charts (daily, weekly, monthly trends)
- Deficiency/excess alerts based on ICMR guidelines
- Meal quality scoring (diversity, processing level, balance)

5. Hydration Management System

- Personalized water requirement calculation
- Context-aware reminders (activity, weather, health status)
- Intake logging with vessel recognition
- Hydration trend analysis and consistency metrics
- Electrolyte balance guidance for intense activity or hot climate

6. Recipe & Cooking Assistant

- Healthy Indian recipe collection with nutritional info
- Step-by-step instructions with traditional and modern methods
- Cooking time and difficulty ratings
- Ingredient substitution suggestions for health or restrictions
- Meal prep and batch cooking guidance
- Traditional festival and special occasion recipes

7. Health Condition Modules

- Diabetes: Carb counting, GI guidance, meal timing, snack suggestions
- Hypertension: Sodium tracking, potassium-rich foods, DASH adaptations
- Weight Goals: Calorie targeting, macro distribution, meal timing strategies
- Pregnancy: Trimester needs, iron/calcium/folate focus, Indian food sources
- Elderly: Easy digestion, nutrient density, hydration focus, texture modifications
- Pediatric: Growth-supporting nutrition, Indian complementary foods, snack ideas

8. Location-Based Nutrition Services

- Nutritionist/dietitian search with credentials and specialties
- Health food store and organic market locator
- Diet-friendly restaurant finder with healthy Indian options
- Pharmacy locator for recommended supplements
- Walking/jogging route mapper for activity complement
- Farmer's market and local produce finder

9. Settings & User Preferences

- Language selection and regional preference management
- Notification customization (meals, water, meal prep, achievements)
- Unit measurement preferences (metric, traditional Indian)
- Data export and backup options
- Account management (profile edit, password change, data deletion)
- App theme and display preferences (dark/light mode, font size)

3.9 FEASIBILITY STUDY

- Technical Feasibility: Confirmed through prototype testing of food recognition (85% accuracy for top 50 Indian foods), meal plan generation, and personalized recommendations.
- Operational Feasibility: User surveys indicated strong interest (78% positive response) in personalized Indian nutrition app, with willingness to pay for premium features.
- Economic Feasibility: Development costs primarily involve human resources (negligible licensing for Firebase/Expo within free tier); monetization strategy includes freemium model and potential partnerships with health insurers.
- Schedule Feasibility: 20-week timeline allows for thorough database development, user testing across demographic groups, and iterative improvements.
- Legal/Regulatory: Nutritional guidance positioned as educational, not medical advice; clear disclaimers implemented; data handling complies with Indian privacy laws.

CHAPTER 04: SYSTEM DESIGN

4.1 DIAGRAMS

System design establishes Nutrimitra's architectural foundation addressing data flow, component interactions, and user experience requirements.

4.1.1 ACTIVITY DIAGRAM

Illustrates user journey from app launch to nutrition actions:

- Onboarding: Language selection Profile setup (health metrics, goals, conditions) Regional preference configuration
- Daily Routine: Morning hydration reminder Breakfast logging (image/manual) Mid-morning snack suggestion Lunch planning
- Weekly Actions: Meal plan generation (Sunday) Grocery list creation Recipe exploration Progress review Adjustment for upcoming events
- Health Condition Flows: Specialized logging paths for diabetic carb counting, hypertensive sodium tracking, etc.
- Emergency Shortcuts: Quick access to water logging during heat exposure, emergency meal suggestions for hypoglycemia
- Data Persistence: Local caching of food database and recent logs; background sync when connectivity available

4.1.2 CLASS DIAGRAM

Models core entities and relationships:

- User: Core profile with health metrics, goals, conditions, preferences, and regional settings
- FoodItem: Nutritional profile per standard Indian serving (katori, cup, piece) with regional variants
- Meal: Time-stamped consumption record with food items, quantities, and calculated nutrients
- MealPlan: Daily/weekly structure with meal slots, nutritional targets, and alternative suggestions
- Recipe: Ingredient list, preparation method, nutritional info per serving, and cultural context
- WaterIntake: Timestamped consumption with vessel type and volume
- HealthCondition: Specific tracking requirements (diabetes: carbs/sugars; hypertension: sodium; etc.)
- Nutritionist: Professional profile with specialties, location, and consultation offerings
- ProgressRecord: Periodic health metrics (weight, waist, BP, glucose) for trend analysis
- FoodLog: Container for meals consumed within specific time periods (daily/weekly)

Services mediating interactions:

- FoodDatabaseService: Handles search, recognition, and nutrient retrieval
- MealPlanningEngine: Generates plans based on user profile and constraints
- NutritionCalculator: Computes macros/micros from food quantities and types
- ReminderEngine: Schedules context-based notifications (meals, water, activity)
- LocalizationService: Manages language switching and culturally appropriate content
- LocationService: Finds nearby nutrition professionals and health outlets
- SyncManager: Handles offline/online data transition with conflict resolution

4.1.3 USE CASE DIAGRAM

Describes system interactions:

- Primary Actor: End User (Individual seeking nutrition management)
- Secondary Actors: Nutrition Professionals, Health Coaches, Family Members (shared goals)
- Core Use Cases:
 - Authenticate & Setup Profile
 - Log Food Intake (Image/Voice/Manual/Text)
 - View Daily Nutrition Summary
 - Generate/Modify Meal Plan
 - Access Healthy Recipes
 - Track Water Consumption
 - Set & Monitor Health Goals
 - Find Local Nutrition Services

- Review Progress Reports
- Adjust Preferences & Language
- Extended Use Cases:
 - Nutritionist: Client management, plan sharing, progress monitoring
 - Family: Shared goal setting, mutual encouragement, meal coordination
 - Coach: Group planning, challenge creation, community support

4.1.4 ENTITY-RELATIONSHIP DIAGRAM

Shows data relationships in Firestore:

- Users (1) (M) Profiles (health goals, conditions, preferences)
- Users (1) (M) FoodLogs (daily collections of meals)
- FoodLogs (1) (M) Meals (breakfast, lunch, dinner, snacks with time stamps)
- Meals (1) (M) FoodItems (specific foods consumed with quantities)
- Users (1) (M) MealPlans (weekly/daily planning documents)
- MealPlans (1) (M) MealSlots (time-specific meal suggestions)
- Users (1) (M) WaterLogs (daily hydration records)
- Users (1) (M) ProgressRecords (periodic health measurements)
- Users (1) (M) RecipeCollections (user-saved and preferred recipes)
- Global (M) FoodDatabase (master Indian food nutritional repository)
- Global (M) RecipeRepository (community-shared healthy Indian recipes)
- Users (M) (M) Nutritionists (professional connections for guidance)

4.1.5 DATA FLOW DIAGRAM

Traces information movement:

- Input Layer:
 - Food image via camera ML Kit recognition Food item identification
 - Manual entry Predictive text search Food database lookup
 - Barcode scan Product lookup Nutritional data retrieval
 - Voice input Speech-to-text Ingredient parsing Food matching
- Processing Layer:
 - Quantity determination (standard serving, user-specified, image-based estimation)
 - Nutrient calculation (per ICMR/NIN standards using food database values)
 - Goal comparison (daily targets vs. actual intake)
 - Deficiency/excess flagging (based on health condition and general guidelines)
 - Meal quality assessment (diversity, whole foods vs. processed, balance)
- Storage Layer:
 - Local cache: Recent food database, user preferences, unsynced logs
 - Firestore: Permanent user profile, food logs, meal plans, progress records
 - Backup: Encrypted export option for user-controlled data preservation
- Output Layer:
 - Dashboard: Visual summary of daily/weekly nutrition status
 - Alerts: Contextual reminders (meals, water, nutrient gaps)
 - Suggestions: Meal alternatives, recipe ideas, shopping lists
 - Reports: Trend analysis, goal progress, health metric correlations
 - External: Shareable summaries, exportable data, professional consultation prep

4.2 DATA DESIGN

Data design is the process of defining the logical structure of the app's database to ensure efficiency and consistency. For Nutrition

4.2.1 SCHEMA DESIGN

Schema design defines the organization of collections and fields in Firestore. Collections and Typical Fields:

1. users/{uid}
 - o name, email, phone, age, gender, height, weight, activityLevel, healthConditions, nutritionalGoals, languagePreference, regi
2. users/{uid}/foodLogs/{yyyy-mm-dd}

- o meals (subcollection with time-stamped entries for breakfast, lunch, dinner, snacks)
- 3. users/{uid}/foodLogs/{yyyy-mm-dd}/meals/{mealId}
 - o foodItems (array of consumed foods with quantities), mealType (breakfast/lunch/dinner/snack), timeStamp, calculatedNutrients
- 4. users/{uid}/waterLogs/{yyyy-mm-dd}
 - o intakeRecords (array of timestamped water consumption with vessel type and volume)
- 5. users/{uid}/progress/{metric}
 - o value, date, notes (for weight, waist, blood pressure, glucose etc.)
- 6. users/{uid}/mealPlans/{weekStartDate}
 - o dailyMeals (array of 7 days with meal slots), nutritionalTargets, groceryList, alternatives
- 7. recipes/{recipeId}
 - o title, ingredients (with quantities), preparationSteps, nutritionalInfoPerServing, prepTime, difficulty, dietaryTags, imageUrl
- 8. foodDatabase/items/{foodId}
 - o name, region, category (breakfast/snack/main/etc.), preparationMethod, servingSize, caloriesPerServing, proteinPerServing
- 9. nutritionists/{nutritionistId}
 - o name, credentials, specialties, location, consultationMode, contactInfo, availability, rating
- 10. healthFoodStores/{storeId}
 - o name, location, productFocus, priceRange, services, contactInfo, rating

4.2.2 DATA INTEGRITY AND CONSTRAINTS

Data integrity ensures that information is accurate, consistent, and validated before being stored. Integrity Rules and Constraints

- Users:

- * email required/format check;
- * age must be positive integer;
- * height and weight must be within realistic ranges;
- * languagePreference must be from predefined list of 10 Indian languages;
- * regionalPreference must be from North/South/East/West/Central Indian or Mixed

- Food Logs:

- * mealType must be one of: breakfast, lunch, dinner, snack;
- * timeStamp must be valid datetime;
- * foodItems array must not be empty when logging a meal;
- * quantities must be positive numbers;
- * calculatedNutrients must be non-negative

- Water Logs:

- * vesselType must be from predefined list (glass, bottle, kulhad, matka, custom);
- * volume must be positive;
- * timeStamp must be valid datetime

- Progress Records:

- * metric must be from predefined health metrics (weight, waist, BP, glucose etc.);
- * value must be within realistic range for the metric;
- * date must not be in future

- Meal Plans:

- * weekStartDate must be a Sunday (start of week);
- * dailyMeals must contain exactly 7 day entries;
- * nutritionalTargets must be positive numbers;
- * groceryList items must have positive quantities

- Recipes:

- * title required;
- * ingredients array must not be empty;
- * preparationSteps array must not be empty;
- * nutritionalInfoPerServing must contain calories, protein, carbs, fat;
- * prepTime must be positive integer (minutes)

- Food Database:

- * name required;
- * region must be from predefined Indian regions;

- * category must be valid meal course;
- * servingSize must be positive;
- * all nutrient values per serving must be non-negative;
- * dietaryTags must be from predefined list
- Firestore Security Rules enforce user-specific read/write permissions, ensuring users can only access their own nutritional data and preventing unauthorized access to others' information.

4.3 ALGORITHM DESIGN

Algorithm design defines the logical flow of the main functionalities within the Nutrimitra application:

1. Food Recognition Algorithm

- o Input: image from camera or gallery
- o Steps: preprocess image ML Kit classification portion estimation based on food type nutrient calculation using food database
- o Output: logged meal with nutritional breakdown

2. Meal Planning Algorithm

- o Input: user profile (health metrics, goals, conditions, preferences)
- o Steps: calculate BMR using Mifflin-St Jeor equation adjust for activity level apply health condition constraints distribute calories
- o Output: weekly meal plan with daily slots, nutritional targets, and shopping list

3. Nutrition Tracking Algorithm

- o Input: logged food items with quantities
- o Steps: retrieve nutrient values per 100g from food database scale to actual quantity consumed sum macros and micros compare to targets
- o Output: nutrient breakdown, goal progress percentage, meal quality score, deficiency/excess alerts

4. Hydration Management Algorithm

- o Input: user weight, activity level, climate data, health conditions
- o Steps: calculate base water need (35ml/kg) apply activity adjustment apply climate adjustment apply health condition adjustments
- o Output: personalized daily goal, reminder schedule, intake log, hydration analytics

5. Recipe Suggestion Algorithm

- o Input: user profile, meal plan requirements, dietary restrictions
- o Steps: filter recipes by nutritional compatibility match to meal timing consider preparation time prioritize user preferences suggest alternatives
- o Output: list of suitable recipes with nutritional info and preparation guidance

6. Health Condition Module Algorithms

- * Diabetes:
 - Carb counting: sum carbohydrates from logged foods compare with targets suggest adjustments
 - GI guidance: label foods by glycemic index recommend low-GI combinations
- * Hypertension:
 - Sodium tracking: sum sodium from logged foods compare with limits suggest low-sodium alternatives
 - Potassium emphasis: highlight potassium-rich foods in suggestions
- * Weight Management:
 - Calorie targeting: calculate deficit/surplus adjust meal plan track progress toward goal

7. Location-Based Services Algorithm

- o Input: user location, service type (nutritionist, store, restaurant)
- o Steps: request location permission query OpenStreetMap Overpass API filter by specialty/services sort by distance render results
- o Output: list of nearby nutritional resources with distance, contact info, and navigation links

4.4 USER INTERFACE DESIGN

Nutrimitra features a clean, accessible interface built with React Native and Expo. The design supports 10 Indian languages, offline mode, and a dark theme.

1. Usability: Quick-add tiles for common meals, large touch targets, inline validation
2. Consistency: Unified color palette inspired by Indian spices, consistent typography, standard iconography
3. Responsiveness: Works across devices; handles offline gracefully with sync indicator
4. Cultural Sensitivity: Incorporates traditional Indian measurement systems, respects dietary restrictions, uses familiar food representations

Key Screens:

- **Login/Sign Up Page**:

- * Purpose: Secure authentication before accessing personal nutrition data

- * Elements: Logo, email/phone input, password input, sign-in button, sign-up link, forgot password link
 - * Behavior: On success Profile Setup or Dashboard; on error validation message
- **Profile Setup Page**:
 - * Purpose: Collect user health information and preferences
 - * Elements: Health metrics (weight/height converter), activity level selector, goal setting, health condition checkboxes, regional
 - * Behavior: Progressive disclosure to prevent overwhelm; completion indicator; save button
- **Dashboard**:
 - * Purpose: Overview of daily nutrition status
 - * Elements: Greeting with name, water intake progress ring, meal logging quick-add tiles, nutrition summary (calories/macros),
 - * Behavior: Tabs for Today/Week/Month view; pull-to-refresh; tapping items navigates to detailed views
- **Food Logger**:
 - * Purpose: Record meal consumption
 - * Elements: Camera button for image recognition, voice input icon, barcode scanner, manual search bar, quick-add tiles for common
 - * Behavior: Image capture ML recognition portion estimation confirmation nutrient display save to log
- **Meal Planner**:
 - * Purpose: View and modify meal plans
 - * Elements: Weekly calendar view, daily meal slots with alternatives, nutritional targets per meal, swap button, grocery list generator
 - * Behavior: Tap slot to see alternatives; long-press to edit; swap with one tap; generate shopping list
- **Nutrition Tracker**:
 - * Purpose: Visualize nutrient intake and progress
 - * Elements: Macronutrient pie chart, micronutrient bar chart, daily goal progress rings, weekly trends, meal quality score, deficiency alerts
 - * Behavior: Tap chart for detailed view; swipe to change date range; tap alert for actionable advice
- **Water Tracker**:
 - * Purpose: Monitor hydration status
 - * Elements: Daily goal progress ring, intake log with timestamps, vessel selector (+1 glass/+250ml quick buttons), reminder status
 - * Behavior: Log intake update progress intelligent rescheduling of next reminder celebration when goal met
- **Recipe Browser**:
 - * Purpose: Discover healthy Indian recipes
 - * Elements: Search bar, filters (meal type, prep time, difficulty, dietary tags), recipe cards (image, title, time, servings, calories),
 - * Behavior: Tap card for full recipe; "Add to Plan" button; share option; save to favorites
- **Health Condition Module** (example: Diabetes):
 - * Purpose: Specialized tracking for condition management
 - * Elements: Carb counter, GI food labels, medication timing suggestions, snack recommendations, trend visualization (daily carb intake)
 - * Behavior: Log meal auto carb calculation alerts if outside target range personalized advice
- **Location Services**:
 - * Purpose: Find nutrition professionals and health resources
 - * Elements: Map view, list view, filter options (specialty, consultation mode), distance display, contact buttons (call, email, WhatsApp)
 - * Behavior: Tap item for details; get directions; initiate contact; save to favorites
- **Settings**:
 - * Purpose: Customize app experience
 - * Elements: Language selector, regional preference, notification controls, measurement system toggle, data export, account management
 - * Behavior: Immediate language change; notification timing adjustment; data backup scheduling; profile edit

4.5 TEST CASE DESIGN

Testing ensures that all modules of Nutrimitra perform as intended, handle user input correctly, and maintain nutritional data acc

4.5.1 AUTHENTICATION MODULE

TC No.	Test Case Name	Description	Test Steps	Expected Result
--------	----------------	-------------	------------	-----------------

TC-A1	Valid Login	Verify login with valid credentials	Enter valid email & password Tap Sign In	Dashboard opens successfully
-------	-------------	-------------------------------------	--	------------------------------

TC-A2	Invalid Login	Invalid password handling	Enter wrong password Tap Sign In	Error message "Invalid credentials"
-------	---------------	---------------------------	----------------------------------	-------------------------------------

TC-A3	Missing Fields	Required fields validation	Leave fields empty Tap Sign In	Show "Email and Password required"
-------	----------------	----------------------------	--------------------------------	------------------------------------

TC-A4	Create Account	Account creation with valid details	Tap Create Account Enter data Submit	New user created and redirected
-------	----------------	-------------------------------------	--------------------------------------	---------------------------------

TC-A5	Duplicate Email	Existing user email	Try to register with same email	Error "Email already in use"
-------	-----------------	---------------------	---------------------------------	------------------------------

TC-A6	Forgot Password	Password reset process	Tap Forgot Password Enter email	Reset link sent successfully
-------	-----------------	------------------------	---------------------------------	------------------------------

TC-A7	Session Persistence	Maintain login after app restart	Login Close app Reopen	User remains signed in
-------	---------------------	----------------------------------	------------------------	------------------------

TC-A8	Logout	Logout behavior	Tap Logout	Redirected to Login screen
-------	--------	-----------------	------------	----------------------------

TC-A9	Biometric Login	Fingerprint/face ID authentication	Enable biometric Login with biometric	Success login with biometric
-------	-----------------	------------------------------------	---------------------------------------	------------------------------

4.5.2 INDIAN FOOD DATABASE & RECOGNITION MODULE

TC No.	Test Case Name	Description	Test Steps	Expected Result
--------	----------------	-------------	------------	-----------------

TC-F1	Food Search	Valid search query	Enter "dal makhani" Search	Shows relevant results with nutrition info
-------	-------------	--------------------	----------------------------	--

TC-F2	Food Search	Invalid search query	Enter "xyz123" Search	Shows "No results found" message
-------	-------------	----------------------	-----------------------	----------------------------------

TC-F3	Image Recognition	Clear food image	Capture well-lit roti image Process	Shows roti with ~80% confidence
-------	-------------------	------------------	-------------------------------------	---------------------------------

TC-F4	Image Recognition	Low light image	Capture dimly lit food image Process	Shows lower confidence or fallback to manual
-------	-------------------	-----------------	--------------------------------------	--

TC-F5	Barcode Scanning	Valid packaged food	Scan barcode of branded product	Shows correct nutrition info per serving
-------	------------------	---------------------	---------------------------------	--

TC-F6	Barcode Scanning	Invalid barcode	Scan random numbers	Shows "Invalid barcode" message
-------	------------------	-----------------	---------------------	---------------------------------

TC-F7	Custom Food Entry	Add new food	Enter name, category, nutrients Save	Food saved in user's custom foods
-------	-------------------	--------------	--------------------------------------	-----------------------------------

TC-F8	Portion Estimation	Bowl food (dal)	Capture image of katori dal	Estimates quantity as 1 katori (~200ml)
-------	--------------------	-----------------	-----------------------------	---

TC-F9	Portion Estimation	Plate food (roti)	Capture image of 2 rotis	Estimates quantity as 2 pieces
-------	--------------------	-------------------	--------------------------	--------------------------------

TC-F10	Offline Capability	Search without internet	Disable network Search food	Uses cached database, shows results
--------	--------------------	-------------------------	-----------------------------	-------------------------------------

4.5.3 MEAL PLANNING ENGINE MODULE

TC No.	Test Case Name	Description	Test Steps	Expected Result
--------	----------------	-------------	------------	-----------------

TC-MP1	Plan Generation	Weight loss goal	Set goal: lose 0.5kg/week Generate plan	Creates plan with calorie deficit
--------	-----------------	------------------	---	-----------------------------------

TC-MP2	Plan Generation	Diabetes condition	Set condition: diabetes Generate plan	Creates plan with carb distribution (45-60%)
--------	-----------------	--------------------	---------------------------------------	--

TC-MP3	Plan Generation	Regional preference	Set preference: South Indian Generate plan	Includes rice-based meals, sambar,
--------	-----------------	---------------------	--	------------------------------------

TC-MP4	Alternative Generation	View meal alternatives	Tap alternative button on meal	Shows nutritionally similar options
--------	------------------------	------------------------	--------------------------------	-------------------------------------

TC-MP5	Swap Functionality	Replace meal with alternative	Select swap Choose alternative	Meal replaced, nutrition updated
--------	--------------------	-------------------------------	--------------------------------	----------------------------------

TC-MP6	Grocery List Generation	Generate list from plan	Tap generate list	Shows consolidated ingredients with quantities
--------	-------------------------	-------------------------	-------------------	--

TC-MP7	Seasonal Adjustment	Set season: summer	Generate plan	Includes lighter meals, hydrating foods
--------	---------------------	--------------------	---------------	---

TC-MP8	Festival Adjustment	Set date: Diwali	Generate plan	Includes festive-specific healthier options
--------	---------------------	------------------	---------------	---

4.5.4 NUTRITION TRACKER & ANALYZER MODULE

TC No.	Test Case Name	Description	Test Steps	Expected Result
--------	----------------	-------------	------------	-----------------

TC-NT1	Meal Logging	Image-based	Log meal via camera Confirm	Shows correct nutrient breakdown
--------	--------------	-------------	-----------------------------	----------------------------------

TC-NT2	Meal Logging	Manual entry	Search and log "2 rotis" Save	Shows accurate calories/carbs
--------	--------------	--------------	-------------------------------	-------------------------------

TC-NT3	Macro Calculation	Log balanced meal	Log dal, rice, veg Check	Shows proper protein/carbs/fat ratio
--------	-------------------	-------------------	--------------------------	--------------------------------------

TC-NT4	Micro Nutrient Tracking	Log iron-rich food	Log spinach Check	Shows iron contribution
--------	-------------------------	--------------------	-------------------	-------------------------

TC-NT5	Goal Comparison	Log meal within goals	Eat lunch within target	Shows progress ring moving toward goal
--------	-----------------	-----------------------	-------------------------	--

TC-NT6	Deficiency Alert	Log low-iron meals	Log several low-iron meals	Shows "Consider adding spinach" suggestion
--------	------------------	--------------------	----------------------------	--

TC-NT7	Excess Warning	Log high-sodium meal	Log pickle-heavy meal	Shows "High sodium detected" alert
--------	----------------	----------------------	-----------------------	------------------------------------

TC-NT8	Meal Quality Score	Log diverse meal	Log thali with 5 groups	Shows high diversity score
--------	--------------------	------------------	-------------------------	----------------------------

TC-NT9	Historical View	Access weekly/monthly trends	Tap weekly view	Shows charts for past 7 days
--------	-----------------	------------------------------	-----------------	------------------------------

TC-NT10	Export Data	Export nutrition report	Tap export Choose format	Generates JSON/CSV/PDF report
---------	-------------	-------------------------	--------------------------	-------------------------------

4.5.5 HYDRATION MANAGEMENT MODULE

TC No.	Test Case Name	Description	Test Steps	Expected Result
--------	----------------	-------------	------------	-----------------

TC-HM1	Personalized Calculation	Set weight: 70kg, active: moderate	Shows daily goal ~2.45L (adjustments applied)
--------	--------------------------	------------------------------------	---

TC-HM2 Intelligent Reminder Schedule Set wake: 7am, sleep: 11pm Shows reminders spaced 2-3 hours apart
 TC-HM3 Vessel Recognition Log "1 glass" (250ml) Save Records 250ml intake
 TC-HM4 Custom Vessel Log "1 kulhad" (150ml) Save Records 150ml intake with vessel type
 TC-HM5 Trend Analysis Log consistent intake for 3 days View trend Shows 100% consistency percentage
 TC-HM6 Snooze Functionality Tap snooze on reminder 10 min delay Reminder reappears after 10 minutes
 TC-HM7 Escalation Ignore multiple reminders After 3 skips Shows urgent/persistent reminder
 TC-HM8 Context Adjustment Log workout Check reminder frequency Increases during/post exercise
 TC-HM9 Export Hydration Log Tap export Choose format Generates JSON/CSV report of intake

4.5.6 RECIPE & COOKING ASSISTANT MODULE

TC No.	Test Case Name	Description	Test Steps	Expected Result
TC-R1	Recipe Search	Search by ingredient	Enter "spinach" Search	Shows recipes containing spinach
TC-R2	Recipe Filter	Filter by difficulty: easy	Apply filters	Shows only easy preparation recipes
TC-R3	Nutritional Info	View recipe details	Open recipe card	Details Shows calories, macros, micros per serving
TC-R4	Ingredient Substitution	View substitution matrix	Open settings	Substitutions Shows grain/fat/protein alternatives
TC-R5	Meal Planning Integration	"Add to today's plan"	Tap button on recipe	Confirms Adds recipe if fits daily targets
TC-R6	Shopping List Generation	Generate list from recipe	Tap shopping list	Recipe Shows all ingredients with quantities
TC-R7	User Recipe Contribution	Add family recipe	Guided entry Save	Recipe saved with nutrition calculation
TC-R8	Cultural Context	View festival recipe	Open Diwali recipe	Details Shows cultural explanation and healthier adaptations

4.5.7 HEALTH CONDITION MODULE (DIABETES EXAMPLE)

TC No.	Test Case Name	Description	Test Steps	Expected Result
TC-DB1	Carb Counting Log	2 rotis (30g carbs)	View tracker	Shows 30g carbs added to daily total
TC-DB2	GI Labeling	View food details	Open brown rice	Info Shows medium GI label
TC-DB3	Meal Distribution Advice	Log heavy carb dinner	Check next day's plan	Shows lighter carb suggestion
TC-DB4	Hypoglycemia Prevention	Log low-carb meal	Check suggestions	Shows fast-acting carb snack rec
TC-DB5	Trend Analysis	View weekly carb trends	Tap trends tab	Shows daily carb distribution chart
TC-DB6	HbA1c Estimation	Log consistent meals	View estimate	Shows estimated HbA1c based on data
TC-DB7	Educational Content	Access carb sources	Tap education	Details Shows Indian carb sources with viz

4.5.8 LOCATION-BASED SERVICES MODULE

TC No.	Test Case Name	Description	Test Steps	Expected Result
TC-LS1	Nutritionist Search	Search for diabetes specialist	Filter	Shows verified RD with diabetes specialty
TC-LS2	Health Food Store Search	Search for organic store	Filter	Shows stores with organic focus
TC-LS3	Restaurant Finder	Search for diet-friendly option	Filter	Shows restaurants with healthy Indian options
TC-LS4	Contact Initiation	Tap call icon on listing	Action	Opens dialer with correct number
TC-LS5	Email Initiative	Tap email icon	Action	Opens email client with pre-filled subject
TC-LS6	Distance Calculation	View listing	Details	Shows distance in km from current location
TC-LS7	Filter Combination	Filter: pediatric + video consult	Apply	Shows matching nutritionists
TC-LS8	Offline Maps	Disable network	Open maps	Shows last cached locations or permission prompt

4.5.9 SETTINGS & PREFERENCES MODULE

TC No.	Test Case Name	Description	Test Steps	Expected Result
TC-ST1	Language Switch	Change language to Hindi	Restart app	Interface loads in Hindi without data loss
TC-ST2	Measurement Toggle	Switch to traditional units	Log meal	Shows katori/cup instead of ml/grams
TC-ST3	Notification Control	Disable meal reminders	Set time	No meal reminders appear at set time
TC-ST4	Data Export	Export profile data	Choose format	Generates JSON/CSV with user data
TC-ST5	Theme Selection	Switch to dark mode	Restart app	Dark theme applied across all screens
TC-ST6	Font Size Increase	Increase font size	View dashboard	Text appears larger and clearer
TC-ST7	Account Deletion	Initiate delete	Confirm Enter password	Account deleted, data removed from Firestore
TC-ST8	Biometric Login Setup	Enable fingerprint	Login with finger	Success login via biometric